Polarization (Griffiths §4.1): essentials and derivations

Griffiths, *Introduction to Electrodynamics* (5th ed.), pp. 166–172. Condensed from my study notes. Equation numbers match the book.

1. Concepts you need first

- **Charge:** two kinds, like repel, unlike attract. Electron charge $-e$, $e = 1.6 \times 10^{-19}$ C.
- **Conductor:** charges free to move through the body. **Dielectric (insulator):** charges bound to atoms or molecules; they only shift a little.
- **Electric field:** $\mathbf{E} = \mathbf{F}/q$, the force per unit positive charge.
- **Coulomb's law and the point-charge field:** $E = \dfrac{1}{4\pi\epsilon_0}\dfrac{q}{r^2}$, directed outward for $q > 0$.
- **Gauss's law:** $\oint \mathbf{E} \cdot d\mathbf{a} = Q_{\text{enc}}/\epsilon_0$.
- **Dipole:** charges $\pm q$ separated by d; $\mathbf{p} = q\mathbf{d}$, from $-$ to $+$. Unit C·m.
- **Vectors:** components $E_{\parallel} = E\cos\theta$, $E_{\perp} = E\sin\theta$; cross product $|\mathbf{A} \times \mathbf{B}| = AB\sin\theta$; dot product $\mathbf{A} \cdot \mathbf{B} = AB\cos\theta$.
- **Polarization:** dipole moment per unit volume, $\mathbf{P}$, unit C/m².

2. Derivations

2.1 Field of a uniformly charged ball (needed for the atom model)

Let a ball of total charge Q and radius a have uniform charge. At radius $s < a$, the charge inside is $Q\,s^3/a^3$, because the charge is spread evenly through the volume. Gauss's law with a sphere of radius s:

$$E \cdot 4\pi s^2 = \frac{Q\,s^3}{a^3\,\epsilon_0} \quad \Longrightarrow \quad E = \frac{1}{4\pi\epsilon_0}\frac{Q\,s}{a^3}.$$

2.2 Induced dipole and polarizability (Eqs. 4.1, 4.2)

Model the atom as a point nucleus $+q$ inside a uniform electron cloud $-q$ of radius a. Put the atom in a field $\mathbf{E}$. The nucleus moves a distance d from the cloud centre until equilibrium. At equilibrium the applied field equals the cloud's pull on the nucleus. The cloud's field at distance d is (2.1 with $s = d$, $Q = -q$, magnitude only):

$$E = \frac{1}{4\pi\epsilon_0}\frac{q\,d}{a^3}.$$

The induced dipole is $p = qd$. So

$$\mathbf{p} = (4\pi\epsilon_0 a^3)\,\mathbf{E} = \alpha\mathbf{E} \quad (4.1), \qquad \alpha = 4\pi\epsilon_0 a^3 = 3\epsilon_0 v \quad (4.2),$$

where $v = \frac{4}{3}\pi a^3$ is the volume of the ball. The model is crude; the book says it is good to a factor of about 4.

2.3 Units of α

p is in C·m and E is in N/C, so $\alpha = p/E$ is in C²·m/N. Table 4.1 quotes $\alpha/4\pi\epsilon_0$ in $10^{-30}\,\mathrm{m}^3$. Since $4\pi\epsilon_0$ has units C²/(N·m²), multiplying by m³ gives C²·m/N as well.

2.4 Molecules (Eq. 4.3)

If α depends on direction, split $\mathbf{E}$ into parallel and perpendicular parts:

$$\mathbf{p} = \alpha_\parallel \mathbf{E}_\parallel + \alpha_\perp \mathbf{E}_\perp .$$

For an asymmetric molecule, each component of $\mathbf{p}$ depends on all components of $\mathbf{E}$. This needs nine constants α_{ij}, the polarizability tensor:

$$p_x = \alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z \quad (\text{and similarly for } p_y, p_z) \quad (4.3).$$

2.5 Torque on a permanent dipole (Eq. 4.4)

In a uniform field, the force on $+q$ is $q\mathbf{E}$ and on $-q$ is $-q\mathbf{E}$. Net force is zero. Take the centre as the origin, so the ends are at $\pm\mathbf{d}/2$:

$$\mathbf{N} = \frac{\mathbf{d}}{2} \times q\mathbf{E} + \left(-\frac{\mathbf{d}}{2}\right) \times (-q\mathbf{E}) = q\,\mathbf{d} \times \mathbf{E} = \mathbf{p} \times \mathbf{E} \quad (4.4).$$

The torque turns $\mathbf{p}$ toward $\mathbf{E}$.

2.6 Potential energy (Eq. 4.6)

Let θ be the angle between $\mathbf{p}$ and $\mathbf{E}$. The torque has size $pE\sin\theta$. Slowly turn the dipole from $90°$ to θ; the work I do against the field is

$$W = \int_{\pi/2}^{\theta} pE\sin\phi\, d\phi = -pE\cos\theta = -\mathbf{p}\cdot\mathbf{E}.$$

Storing this work as energy gives $U = -\mathbf{p}\cdot\mathbf{E}$, with zero at $90°$. This holds for a permanent dipole of fixed size (book footnote 3). An induced dipole also depends on E, so the full energy is not just this.

2.7 Force in a non-uniform field (Eq. 4.5)

Let the field at the positive end be $\mathbf{E}(\mathbf{r} + \mathbf{d}/2)$ and at the negative end $\mathbf{E}(\mathbf{r} - \mathbf{d}/2)$. For a short dipole, a first-order change along $\mathbf{d}$ gives

$$\Delta\mathbf{E} \approx (\mathbf{d}\cdot\nabla)\mathbf{E}, \qquad \mathbf{F} = q\,\Delta\mathbf{E} = (\mathbf{p}\cdot\nabla)\mathbf{E} \quad (4.5).$$

Here $\nabla = \hat{x}\,\partial_x + \hat{y}\,\partial_y + \hat{z}\,\partial_z$, the partial derivatives. The field must change noticeably over one molecule for this force to matter, so it is usually small in dielectrics.

2.8 Polarization of many dipoles

If each dipole is $\mathbf{p}$ and there are N per unit volume, all pointing the same way:

$$\mathbf{P} = \frac{\text{total dipole moment}}{\text{volume}} = N\mathbf{p} \quad (\text{unit C/m}^2).$$

2.9 Why alignment is never complete

A permanent dipole in a field has energy $-pE\cos\theta$. Thermal motion gives each molecule energy of order $k_B T$ (≈ 0.026 eV at room temperature). If $pE \ll k_B T$, thermal motion wins, and the dipoles point in all directions. For ordinary fields pE is very small, so the net alignment is weak.

3. What you must be able to do

- Derive $\mathbf{p} = \alpha\mathbf{E}$ with $\alpha = 4\pi\epsilon_0 a^3 = 3\epsilon_0 v$ (Section 2.1–2.2).
- State the units of α and check them (Section 2.3).
- Derive $\mathbf{N} = \mathbf{p} \times \mathbf{E}$ and $U = -\mathbf{p}\cdot\mathbf{E}$ (Sections 2.5–2.6).
- Write the force in a non-uniform field as $(\mathbf{p}\cdot\nabla)\mathbf{E}$ and say what ∇ means (Section 2.7).
- Give $\mathbf{P} = N\mathbf{p}$ and its unit (Section 2.8).
- Explain qualitatively why thermal motion limits alignment (Section 2.9).